behaviour is analysed by the **Health Engine**, which evaluates compressor health, identifies abnormal operating conditions, and generates maintenance recommendations.

Unlike conventional monitoring systems, the proposed Digital Twin not only visualizes the compressor's operating condition but also assists operators in understanding the equipment's performance and making informed maintenance decisions.

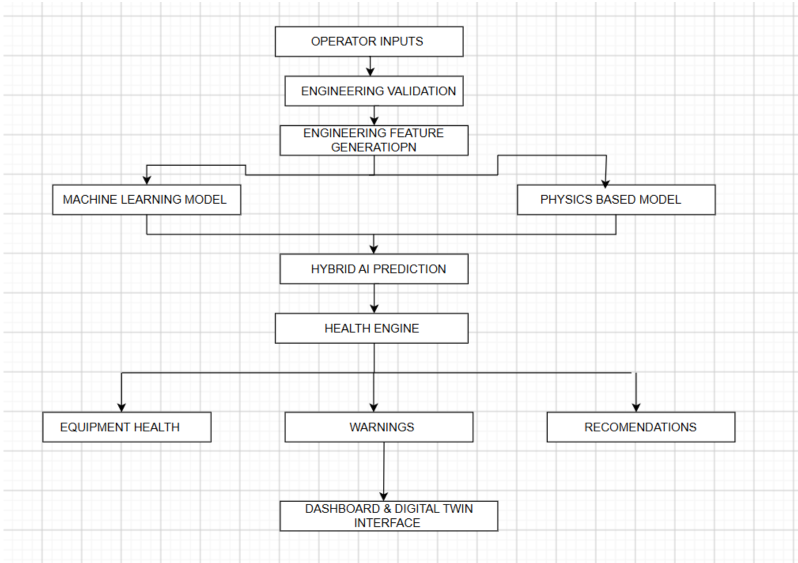

Fig. 3 Overall Architecture of the Proposed Intelligent Digital Twin Framework

1.6 OBJECTIVES OF THE PROJECT

The primary objective of this project is to develop an **Intelligent Digital Twin Framework** capable of evaluating the health of a booster compressor operating in a CNG daughter station. The framework is designed to combine engineering principles with Artificial Intelligence techniques to support intelligent condition monitoring and maintenance decision-making.

The specific objectives of the project are as follows:

- **To develop a Digital Twin framework** for representing the operating behaviour of a booster compressor in a CNG daughter station.
- **To acquire and process compressor operating data** including suction pressure, discharge pressure, compressor delivery flow, voltage, current, and operating hours (HMR).
- **To perform engineering feature generation** by deriving parameters such as pressure ratio, pressure difference, electrical load, compressor capacity utilization, and load index.
- **To develop a Machine Learning model** for estimating the compressor power consumption using historical operating data.
- **To develop a Physics-Based Power Prediction Model** using engineering relationships between compressor operating parameters.